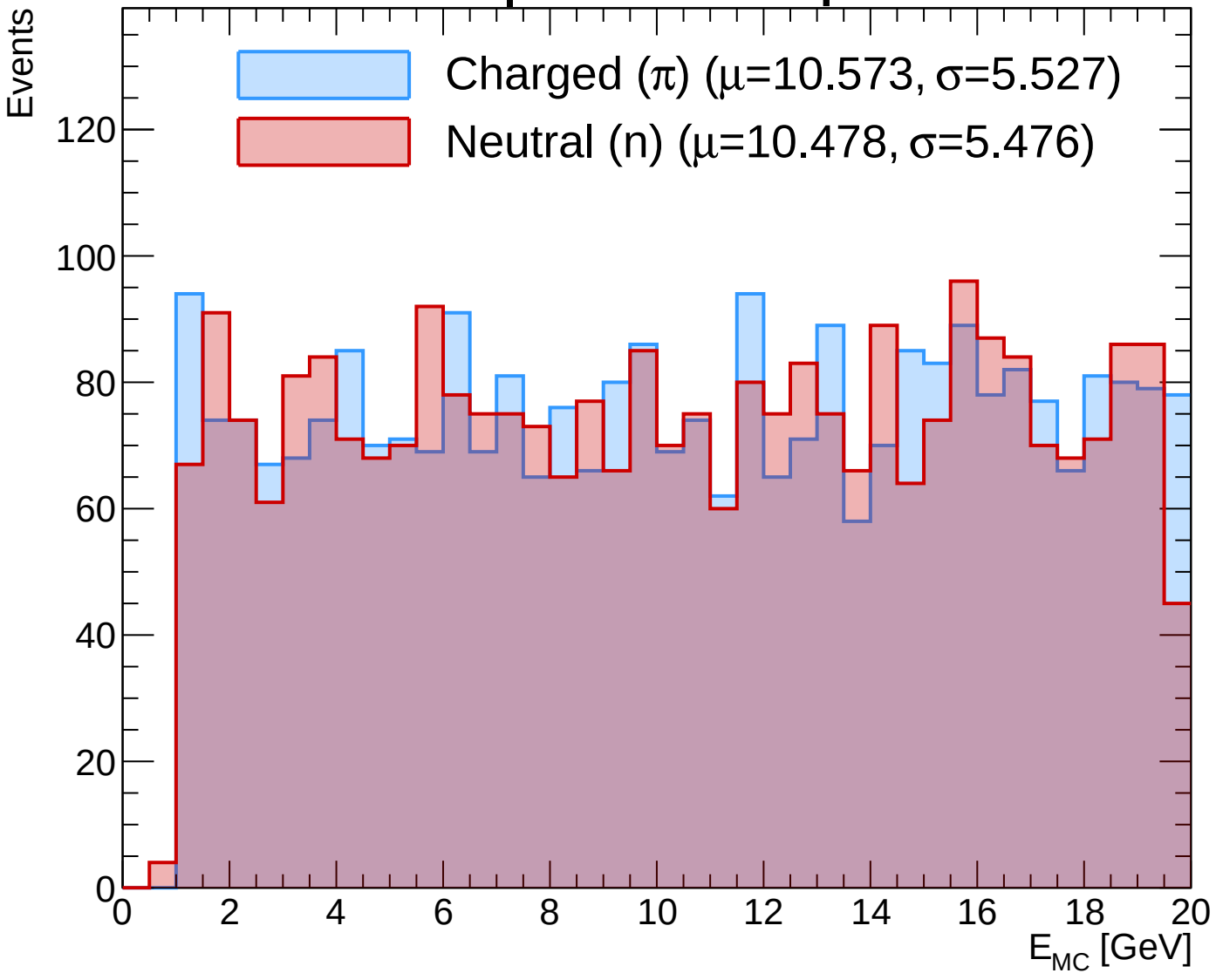
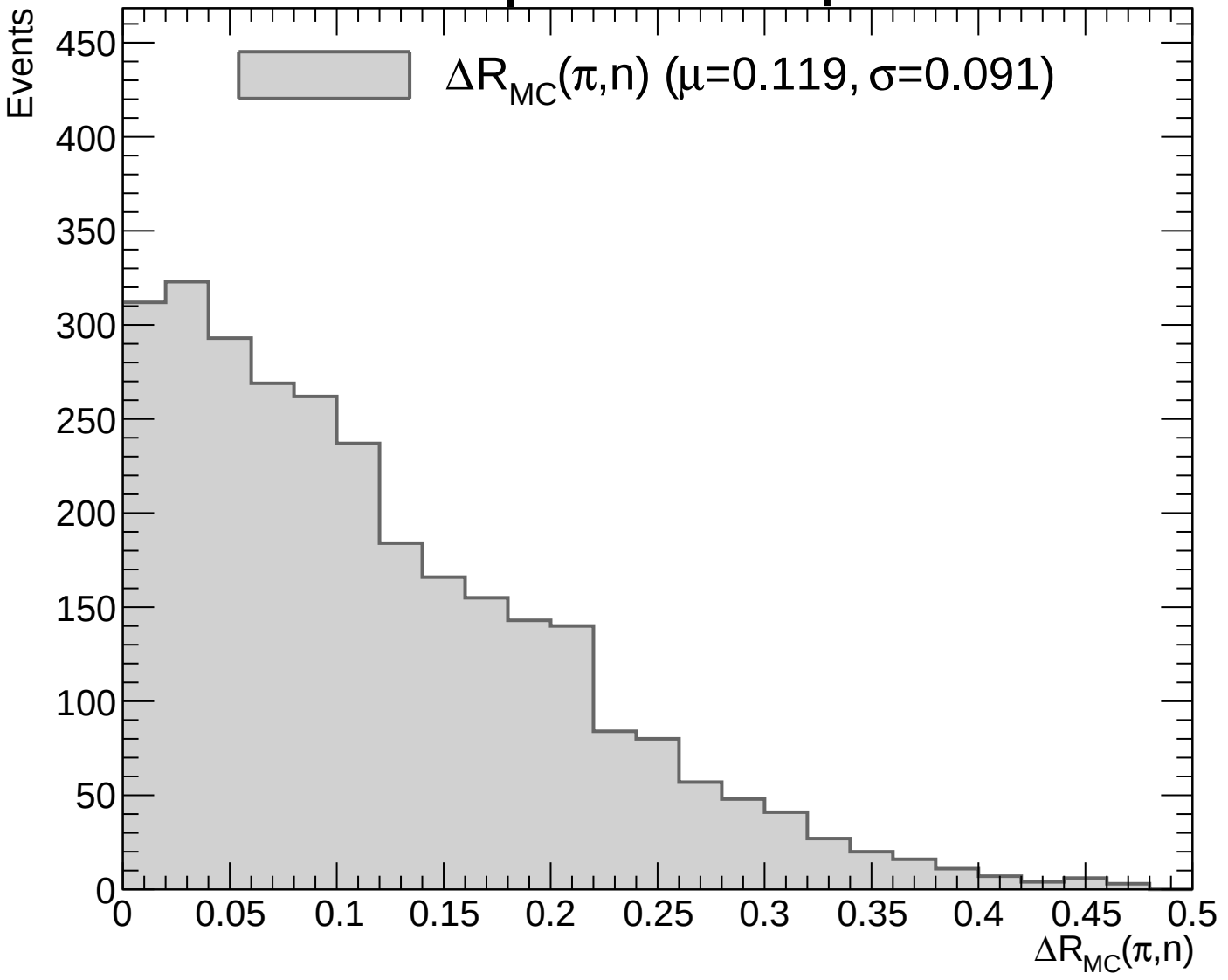
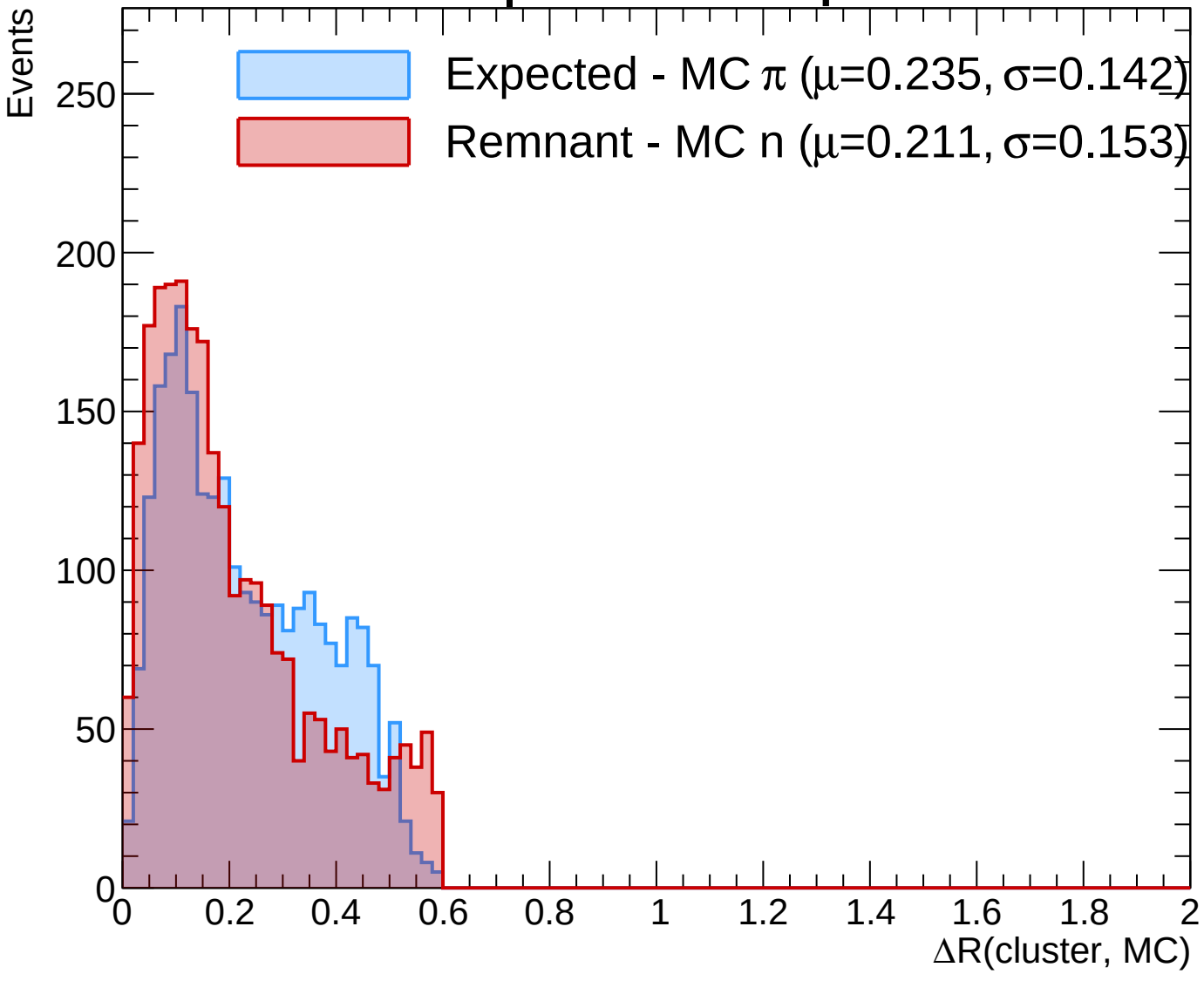
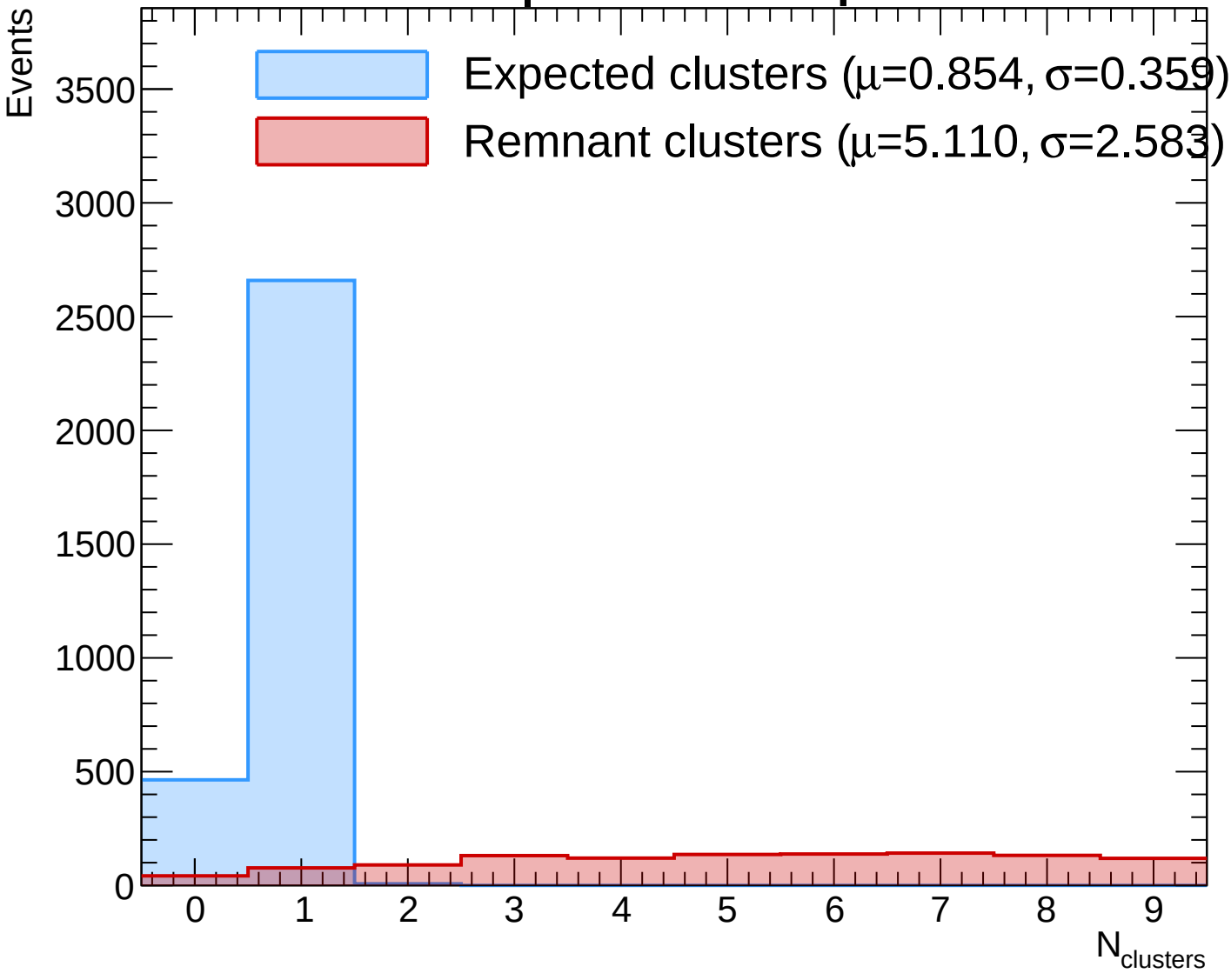
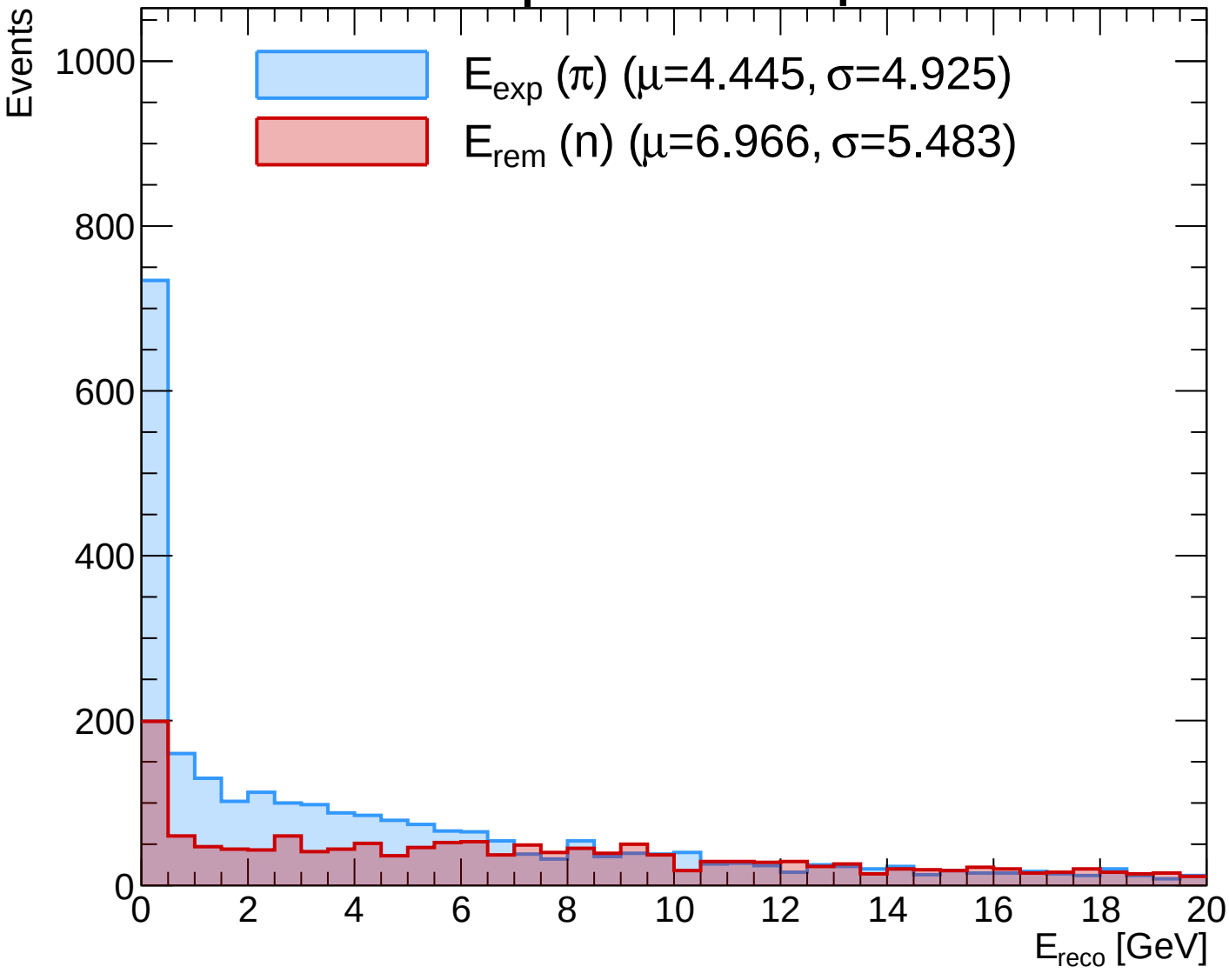
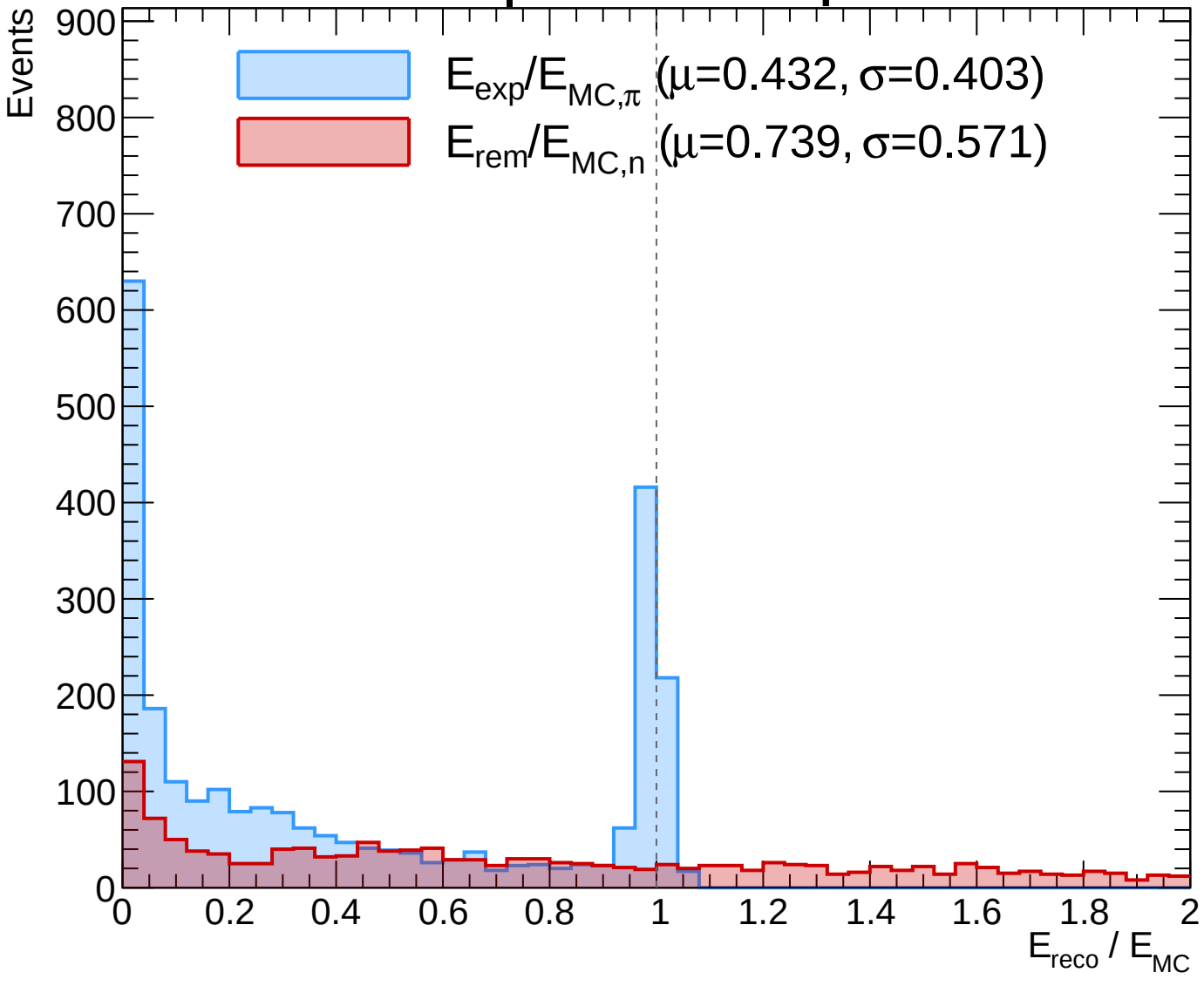

ePIC Benchmark | π⁺+n N Endcap HCal
Charged (π) (μ=10.573, σ=5.527)
Neutral (n) (μ=10.478, σ=5.476)
Events
0
20
40
60
80
100
120
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]

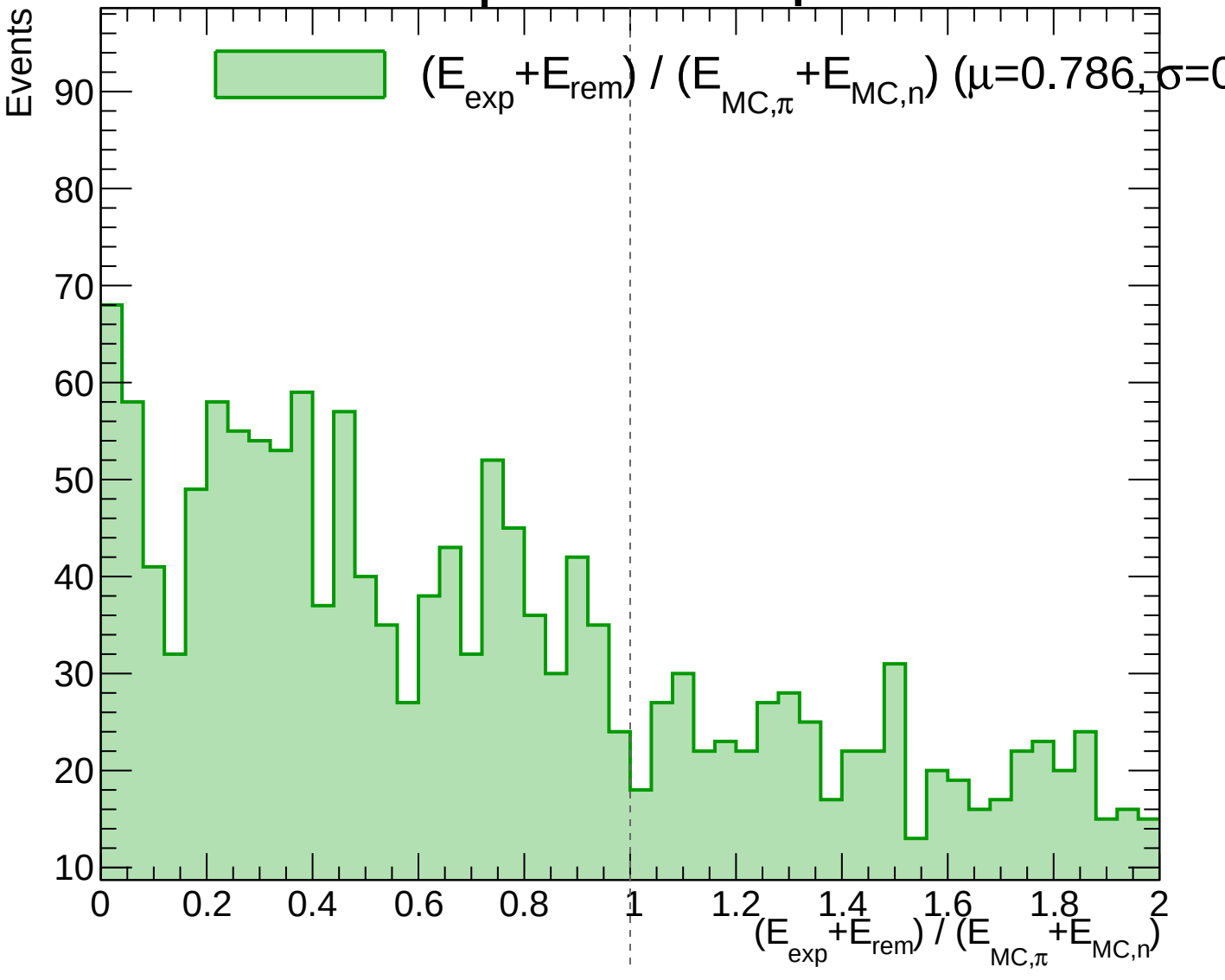

ePIC Benchmark | π⁺+n N Endcap HCal
Events
ΔR_MC(π,n) (μ=0.119, σ=0.091)
450
400
350
300
250
200
150
100
50
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)

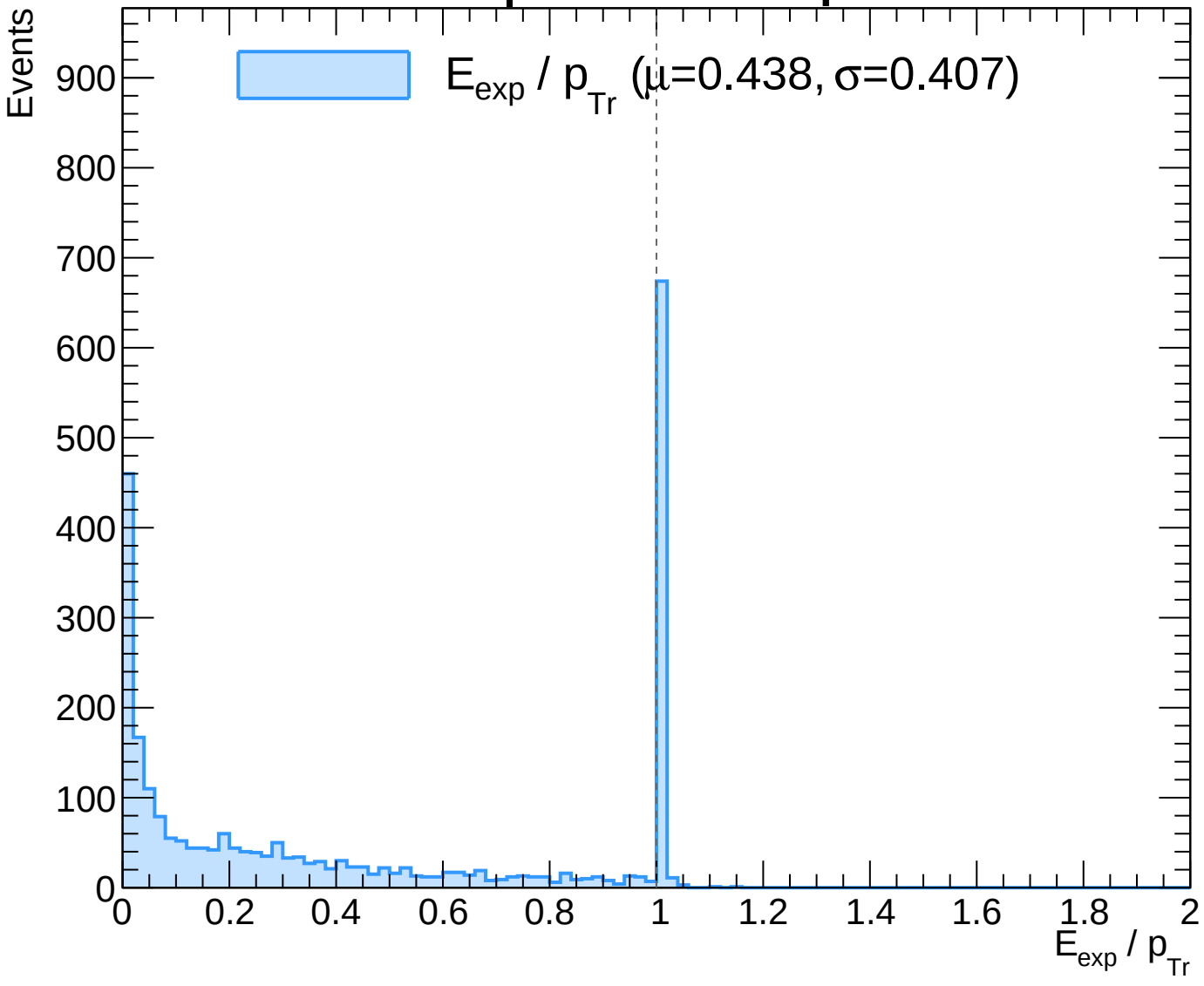

ePIC Benchmark | π⁺+n N Endcap HCal
Events
Expected - MC π (μ=0.235, σ=0.142)
Remnant - MC n (μ=0.211, σ=0.153)
0
50
100
150
200
250
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΔR(cluster, MC)

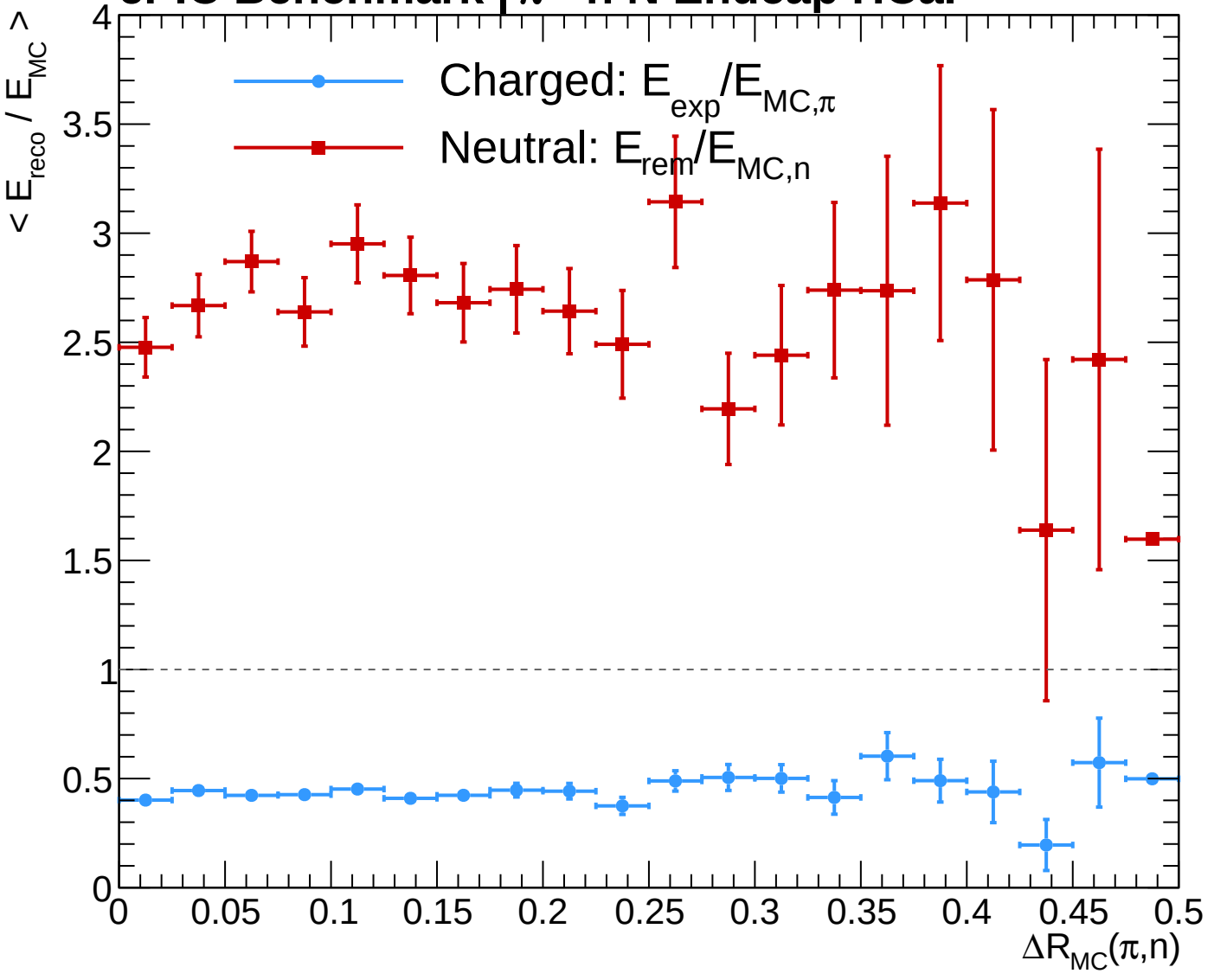

ePIC Benchmark | π⁺+n N Endcap HCal
Events
Expected clusters (μ=0.854, σ=0.359)
Remnant clusters (μ=5.110, σ=2.583)
3500
3000
2500
2000
1500
1000
500
0
0
1
2
3
4
5
6
7
8
9
N_clusters

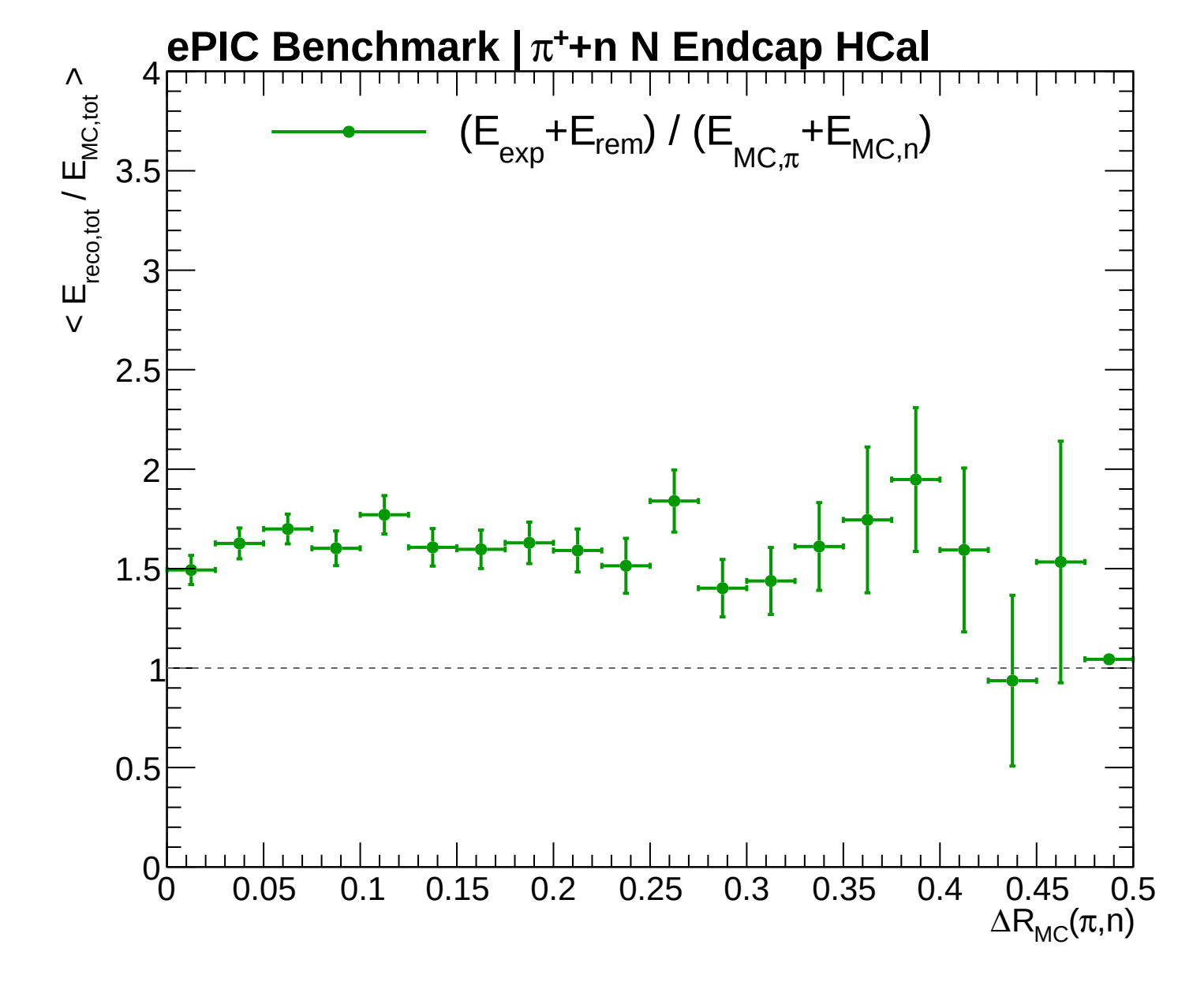

ePIC Benchmark | π⁺+n N Endcap HCal
Events
1000
800
600
400
200
0
0
2
4
6
8
10
12
14
16
18
20
E_reco [GeV]
E_exp (π) (μ=4.445, σ=4.925)
E_rem (n) (μ=6.966, σ=5.483)

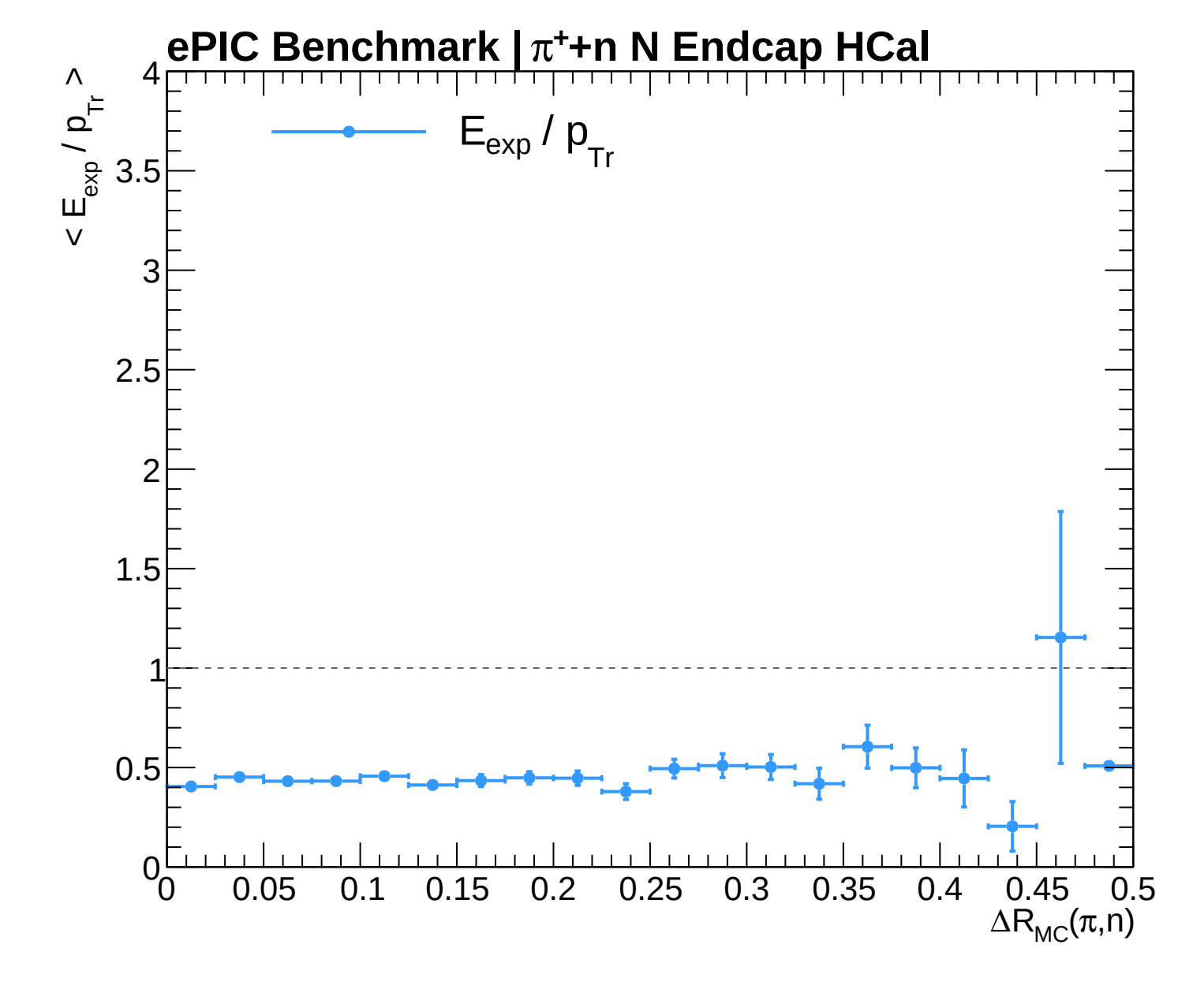

ePIC Benchmark | π⁺+n N Endcap HCal
Events
E_exp/E_MC,π (μ=0.432, σ=0.403)
E_rem/E_MC,n (μ=0.739, σ=0.571)
E_reco / E_MC
0
100
200
300
400
500
600
700
800
900
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

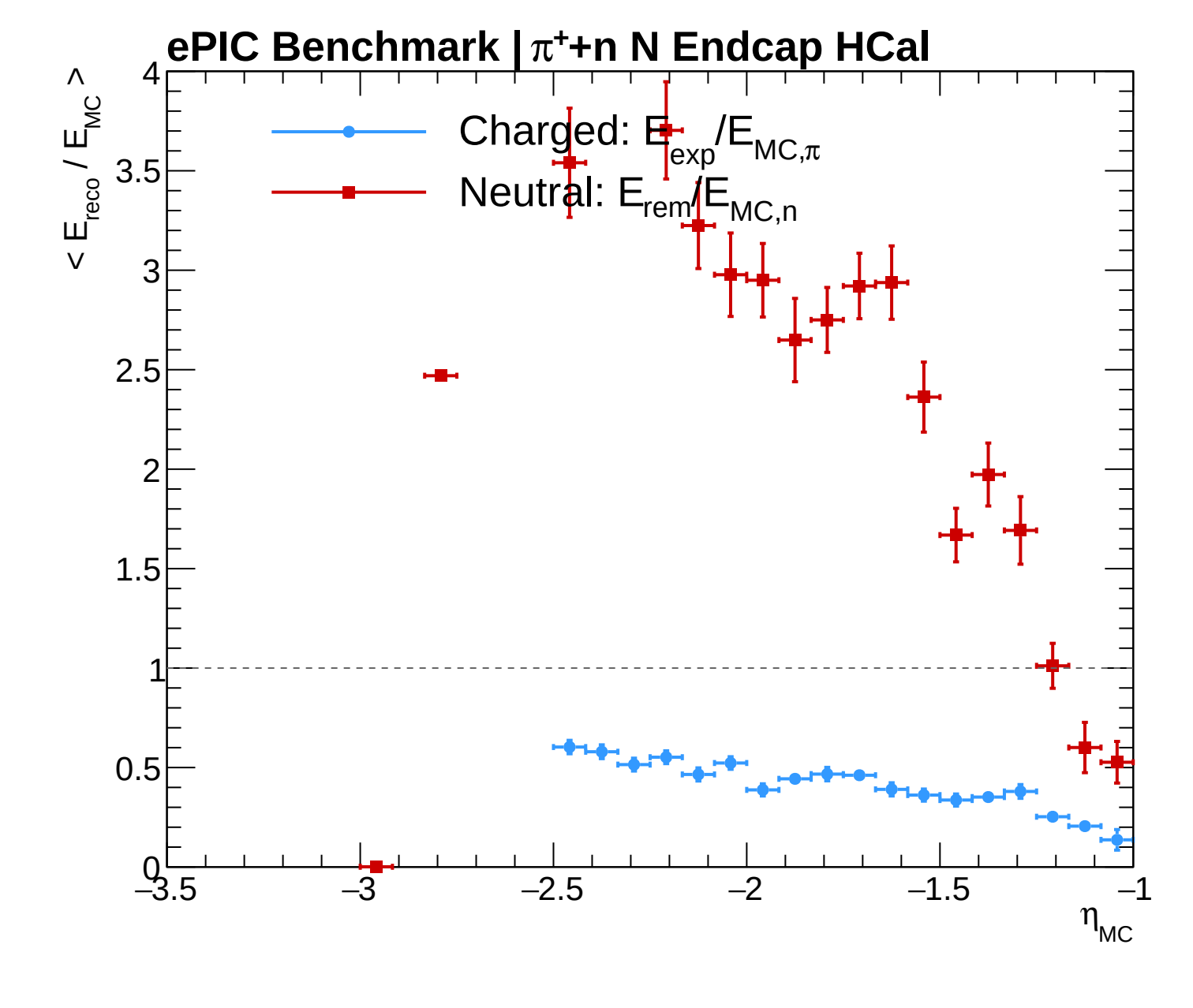

ePIC Benchmark | π⁺+n N Endcap HCal
Events
(E_exp+E_rem) / (E_MC,π+E_MC,n) (μ=0.786, σ=0
(E_exp+E_rem) / (E_MC,π+E_MC,n)
10
20
30
40
50
60
70
80
90
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

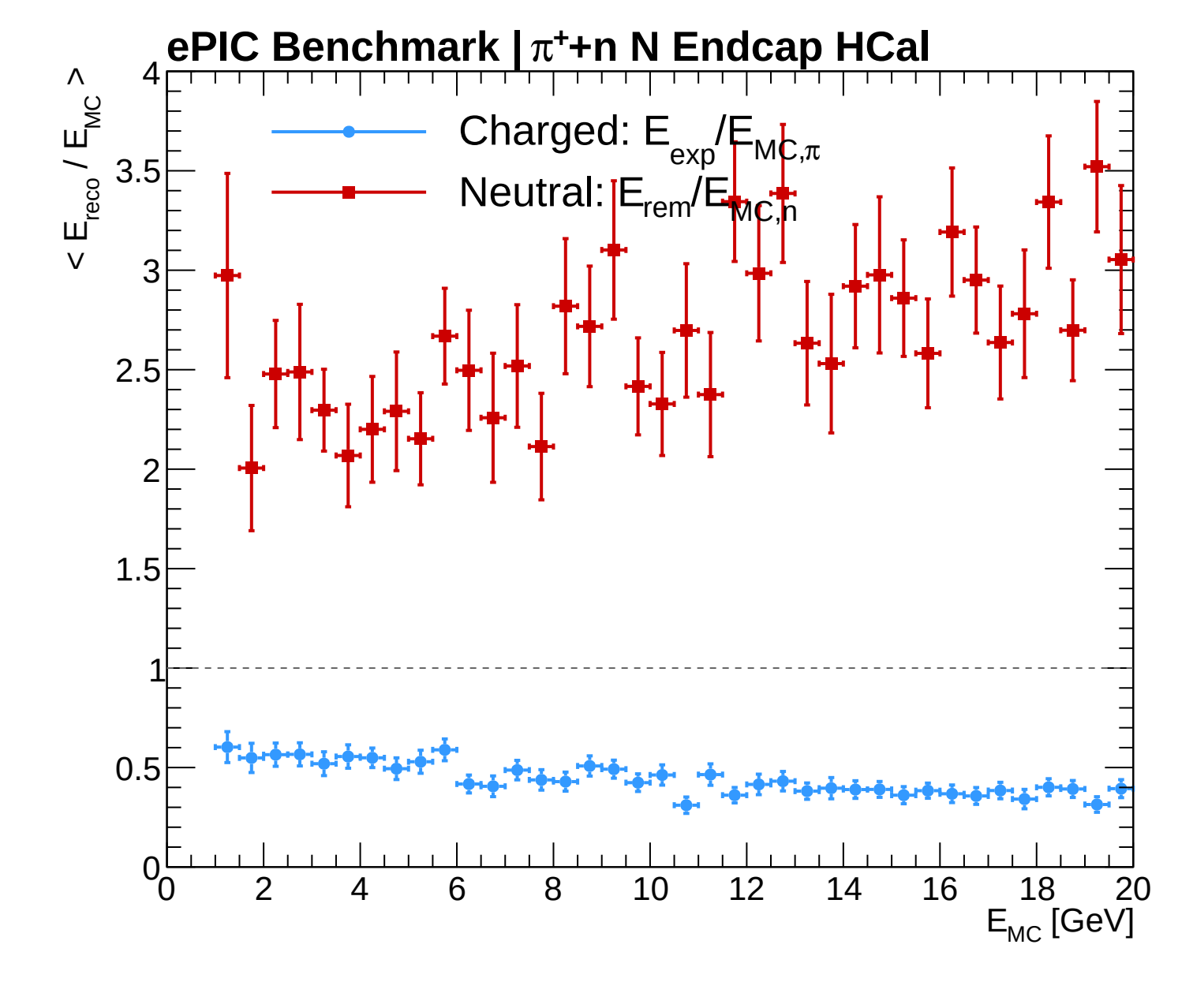

ePIC Benchmark | π⁺+n N Endcap HCal
Events
E_exp / p_Tr (μ=0.438, σ=0.407)
900
800
700
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

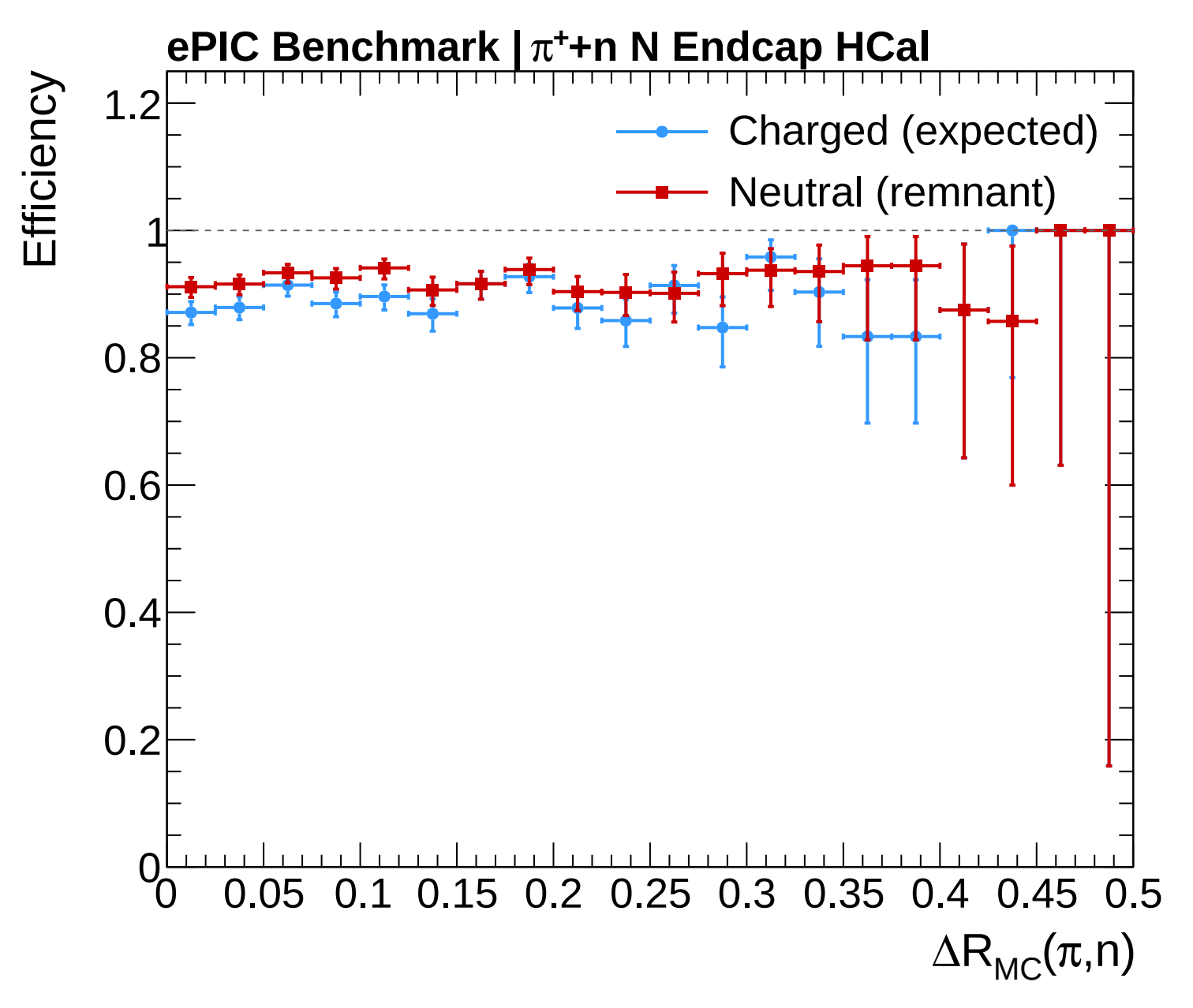

ePIC Benchmark | π⁺+n N Endcap HCal
Charged: E_exp/E_MC,π
Neutral: E_rem/E_MC,n
< E_reco / E_MC >
ΔR_MC(π,n)
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
0.5
1
1.5
2
2.5
3
3.5
4

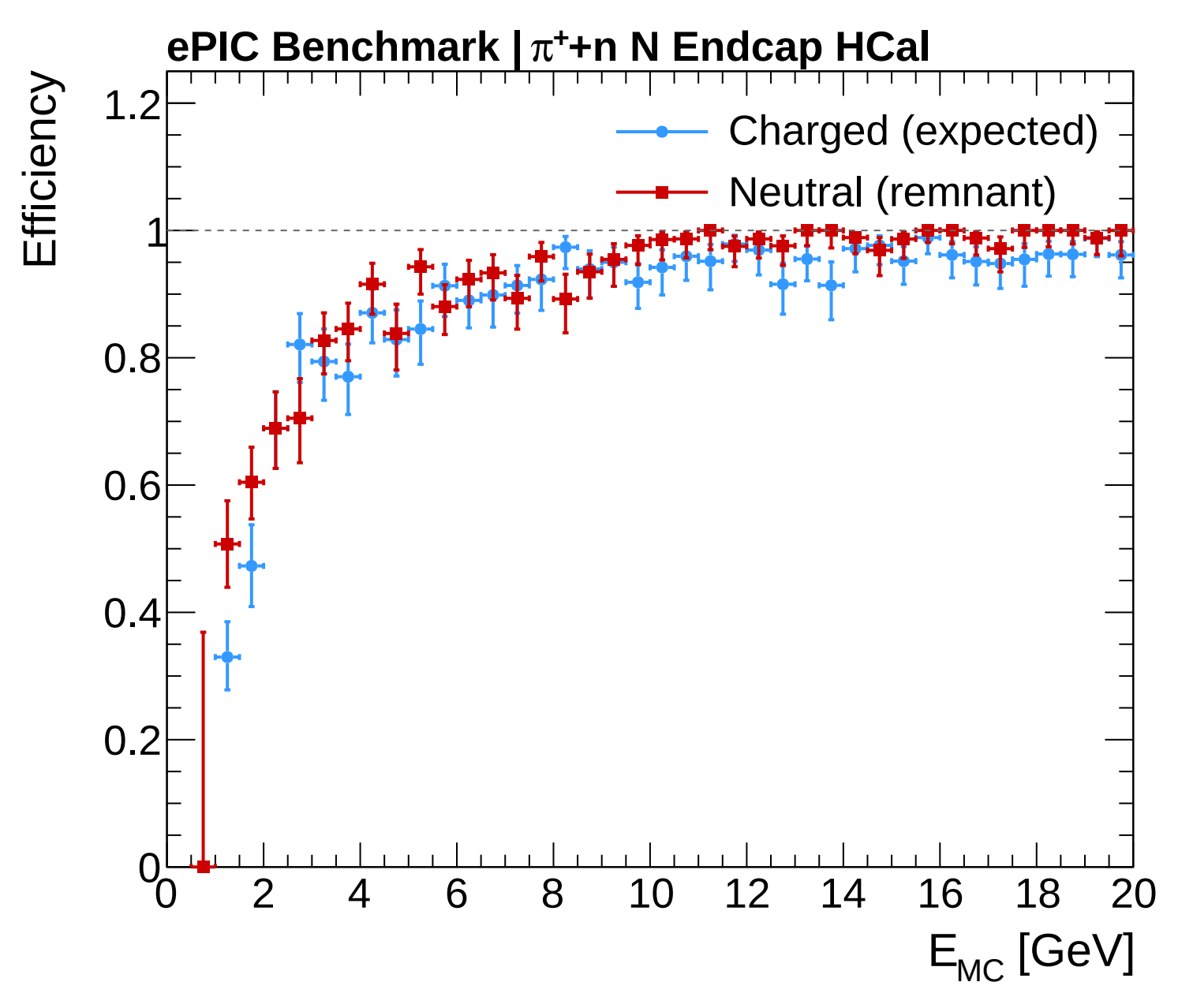

ePIC Benchmark | π⁺+n N Endcap HCal
(E_exp+E_rem) / (E_MC,π+E_MC,n)
< E_reco,tot / E_MC,tot >
ΔR_MC(π,n)
0
0.5
1
1.5
2
2.5
3
3.5
4
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

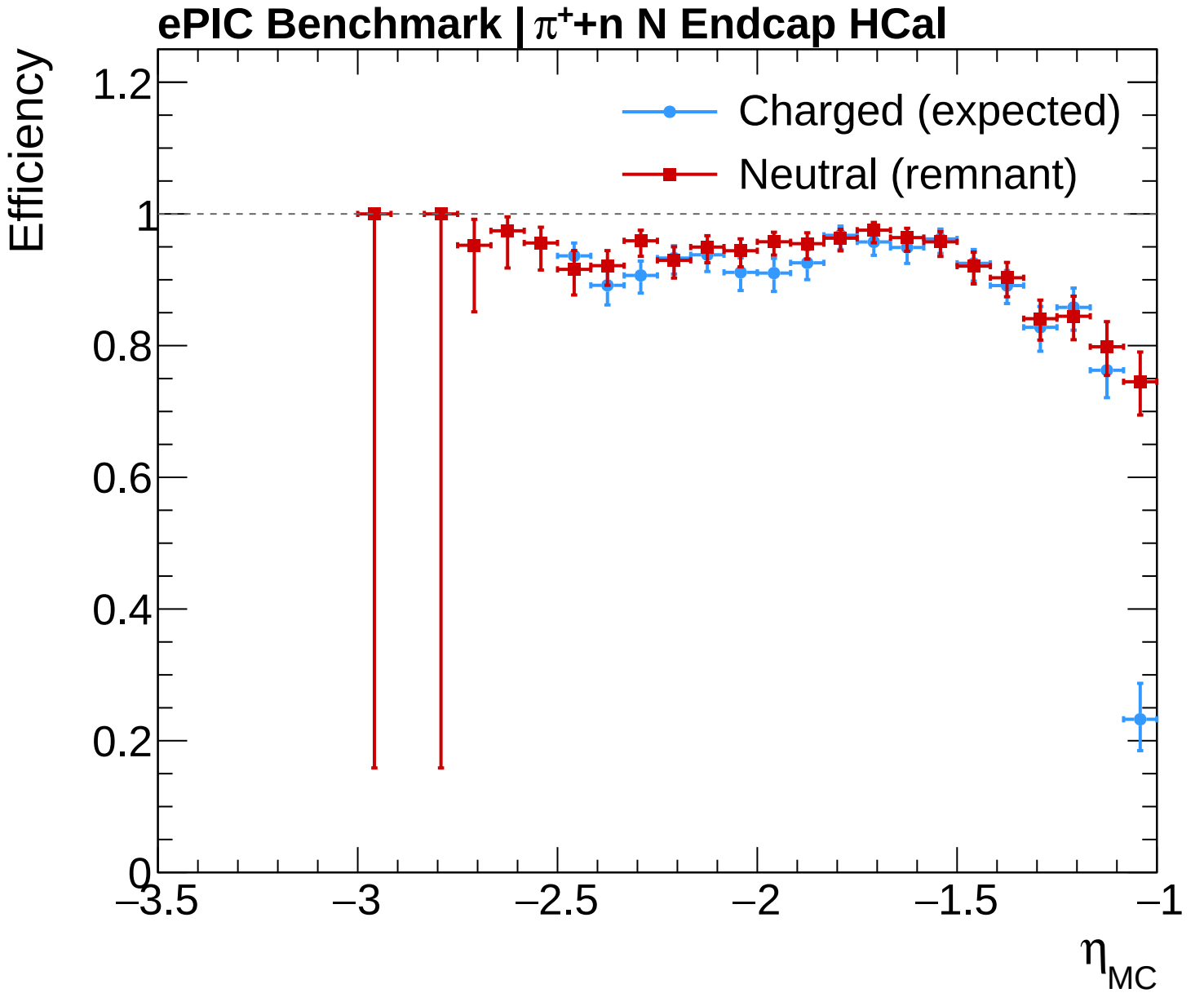

ePIC Benchmark | π⁺+n N Endcap HCal
< E_exp / p_Tr >
E_exp / p_Tr
4
3.5
3
2.5
2
1.5
1
0.5
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)


ePIC Benchmark | π⁺+n N Endcap HCal
Charged: E_exp /E_MC,π
Neutral: E_rem /E_MC,n
< E_reco / E_MC >
η_MC
4
3.5
3
2.5
2
1.5
1
0.5
0
–3.5
–3
–2.5
–2
–1.5
–1


ePIC Benchmark | π⁺+n N Endcap HCal
< E_reco / E_MC >
Charged: E_exp/E_MC,π
Neutral: E_rem/E_MC,n
E_MC [GeV]
0
0.5
1
1.5
2
2.5
3
3.5
4
0
2
4
6
8
10
12
14
16
18
20


ePIC Benchmark | π⁺+n N Endcap HCal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)


ePIC Benchmark | π⁺+n N Endcap HCal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]


ePIC Benchmark | π⁺+n N Endcap HCal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
−3.5
−3
−2.5
−2
−1.5
−1
η_MC